TxBench-PP: Analyzing AI Agent Performance on Preclinical Pharmacology

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ABSTRACT

Artificial intelligence (AI) agents promise to accelerate drug discovery by compressing interpretation and decision-making loops, but practical deployment requires trusted evaluation on realistic program decisions. We introduce TherapeuticsBench Preclinical Pharmacology (TxBench-PP), a verifiable benchmark for small-molecule preclinical pharmacology and the first focused slice of a broader TherapeuticsBench effort across drug-discovery stages and therapeutic modalities. TxBench-PP tests whether agents can recover accurate conclusions from real-world assay data rather than memorized facts from literature. The benchmark contains 100 evaluations indexed by program stage, assay type, and task structure, spanning mechanism-of-action (MoA) and pharmacodynamic (PD) reasoning, compound-target engagement, causal target validation, developability and safety, and translational efficacy. Agents receive realistic workflow snapshots, inspect files in a coding environment, and return structured answers graded deterministically. Across 16 model–harness configurations, comprising 11 models and 4,800 trajectories, no system reliably recovered preclinical pharmacology decisions. The strongest configuration, Claude Opus 4.8 / Pi, passed 59.3% of trajectories, followed by GPT-5.5 / Pi at 55.3%.

TxBench-PP: Overview

TxBench-PP is a focused TherapeuticsBench benchmark for small-molecule preclinical pharmacology. Each evaluation supplies realistic workflow data from a local preclinical decision point and asks whether an agent can recover the result supported by the provided data.

TherapeuticsBench is the broader roadmap across drug-discovery stages and therapeutic modalities. The schematic below contextualizes TxBench-PP with future work.

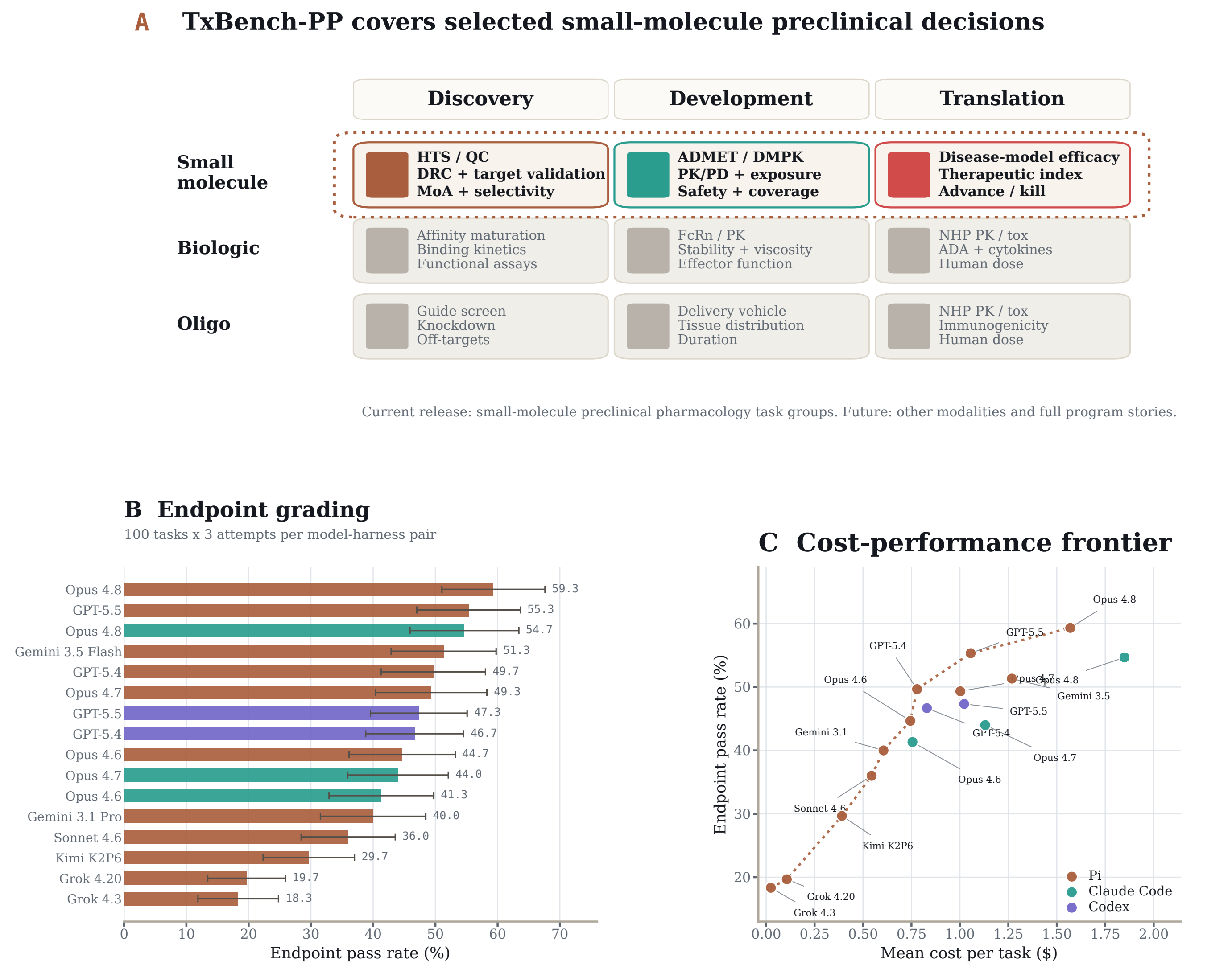


Figure 1: TxBench-PP scope and topline performance. A, TxBench-PP places local small-molecule preclinical pharmacology decisions within the broader TherapeuticsBench roadmap. The dotted boundary marks the task groups in this release; biologics, oligonucleotides, and full program histories are out of scope. **B**, Endpoint pass rates across 16 model–harness configurations, with 95% confidence intervals computed over task-level pass rates. **C**, Mean cost per task against endpoint pass rate.

Introduction

While experiments are rate-limited by natural processes, human decisions and organizational consensus often make up significant components of program timelines in drug discovery. Agents in drug discovery promise to accelerate discovery, development, and translation by compressing these interpretation and decision-making loops. Specific knowledge from constituent steps of the drug-development pipeline can be encoded into advanced models through post-training, tool use, and domain-specific scaffolding, then deployed into practical data interpretation workflows.

However, the practical use of agentic systems in industrial workflows requires standard and trusted methods of evaluating performance. This is especially challenging because the drug-discovery ecosystem is a sprawling landscape of assay categories, development stages, therapeutic modalities, and decision types. Benchmarks must therefore measure realistic tasks while providing focused treatment of the many local scientific judgments that tile the biotech ecosystem.

Preclinical small-molecule pharmacology is one important piece of this landscape. Potency, target engagement, mechanism-of-action evidence, pharmacodynamic response, exposure, safety, and in vivo efficacy are converted into practical decisions about whether to advance, hold, repeat, or kill a candidate. These decisions sit upstream of costly medicinal chemistry, animal studies, and human experiments. They are easy to get wrong and often depend on fragmented expert judgment.

We introduce TxBench-PP, a verifiable benchmark for evaluating AI agents on small-molecule preclinical pharmacology. TxBench-PP is best understood as a focused benchmark within the broader TherapeuticsBench roadmap rather than a complete evaluation of drug discovery holistically. Each evaluation supplies realistic workflow artifacts from a local preclinical decision point and asks the agent to return a structured, deterministically gradable answer. Importantly, the benchmark tests if agents can recover decisions from supplied data instead of leaning on well-understood mechanisms and literature knowledge, with many tasks intentionally designed to penalize systems that do this.

TxBench-PP contains 100 evaluations indexed by program stage, assay type, and task structure. Tasks span mechanism-of-action and pharmacodynamic reasoning, compound-target engagement, causal target validation, developability and safety, and translational efficacy. Other parts of the therapeutic ecosystem, including clinical program stages and non-small-molecule modalities, are left to future TherapeuticsBench efforts. The benchmark offers a measuring stick by which scientists can evaluate agentic systems on local preclinical pharmacology decisions toward practical deployment in drug-discovery programs.

Benchmark Construction

Agents are evaluated on realistic preclinical pharmacology decisions over 100 evaluations spanning eight program stages: disease and model context, screening and hit confirmation, drug response/pharmacogenomics, causal target validation, MoA/PD, compound-target characterization, developability/safety/pharmacokinetics (PK), and translational efficacy. Human genetic target-support tasks are reserved for future benchmarking work. Across the benchmark, evaluations draw on assay types including drug-response screens, genetic perturbation, transcriptomic and proteomic profiling, chemoproteomic and live-cell target engagement, imaging, drug metabolism and pharmacokinetics (DMPK) and absorption, distribution, metabolism, excretion, and toxicity (ADMET) panels, toxicogenomics, and in vivo efficacy or safety studies.

The benchmark is built around recurring local decisions in small-molecule programs. Candidate tasks begin as specific analysis points where assay data might be turned into decisions. Examples include identifying a pathway mechanism from dose-resolved phosphoproteomics, separating true chemoproteomic target engagement from complex contaminants, and interpreting in vivo safety data.

Each evaluation includes the data artifacts needed for that decision, metadata needed to interpret the files, a task description, and a structured answer. Task context is calibrated to approximate what a scientist would know at the point of analysis. We avoid prescribing a method unless the method is part of the scientific decision. The intended answer is a result supported by the supplied data, not some fact or mechanism well understood in the literature. Many tasks are designed to intentionally trap systems that overfit on trained knowledge without empirical exploration.

Candidate evaluations are refined through manual review and model trajectory inspection. We remove tasks when the answer can be recovered by an obvious shortcut, when the prompt over-specifies the analysis, or when plausible analysis choices produce answers that break grading assumptions. The retained tasks are tagged by program stage, assay type, and task structure.

Grading uses deterministic functions over structured final answers. Depending on the task, graders check label sets, numerical tolerances, rank-order relationships, categorical choices, or all required fields. Manual trajectory inspection is part of benchmark construction. We track decision-blocking gaps that should change the program decision if recognized: tox-species mismatch, insufficient free-drug coverage, hook effects, contaminated chemoproteomic signals, survivor bias in pathology sampling, and related failure points. These annotations are not exposed to agents; they are used to interpret failures after endpoint grading and to guide future benchmark updates.

Benchmark Anatomy

TxBench-PP is organized around the anatomy of a practical preclinical pharmacology decision. Each evaluation is tagged along three axes: where the decision sits in a small-molecule program, the assay type that generated the data, and the structure of the task.

The program-stage axis follows the labels shown in Figure 2: disease and model context, screening and hit confirmation, drug response/pharmacogenomics, human genetic target support, causal target validation, MoA/PD, compound-target characterization, developability/safety/PK, and translational efficacy. Human genetic target support is reserved for future benchmarking work because adequately measuring this category requires a separate focused benchmark. The order gives readers context for task progression; it does not imply that every small-molecule program proceeds through this strict sequence in practice.

The assay and task axes describe the actual work required. Assay labels collapse source kits into measurement families such as drug-response screens, protein or post-translational modification (PTM) measurements, genetic perturbation, target engagement, molecular state, safety profiling, and in vivo experiments. Task labels describe broad families of decisions: mechanism inference, quality control (QC), dependency calls, differential response, safety assessment, imaging, hit prioritization, and related judgments.

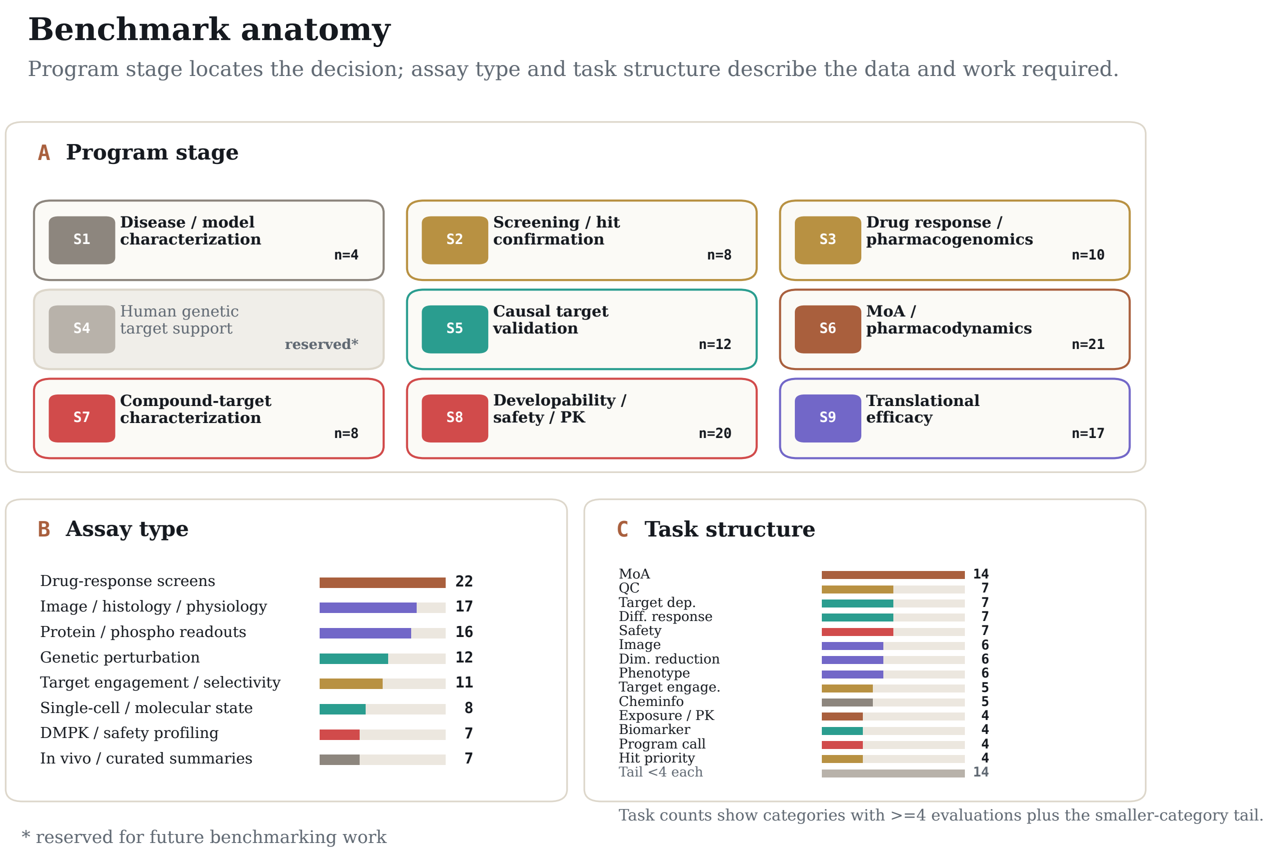


Figure 2: Benchmark anatomy. **A**, Program stage labels place evaluations along a roughly ordered small-molecule evidence spine; the asterisk marks a stage reserved for future benchmarking work. **B**, Assay-type labels summarize the measurement families represented in the source data. **C**, Task-structure labels describe the operation or judgment required for a passing answer.

Results

No agent reliably recovers preclinical pharmacology decisions

We evaluated 16 model–harness configurations, comprising 11 models across three agent harnesses, on 100 preclinical pharmacology tasks. Each configuration was run three independent times per task, yielding 4,800 agent trajectories. A trajectory was scored as passing only if its submitted answer passed every task-specific grader. We report endpoint pass rate as the fraction of passing trajectories, with 95% confidence intervals computed across tasks.

Performance at the top was clustered rather than clearly separated. The strongest configuration, Claude Opus 4.8 on Pi, achieved a pass rate of 59.3% (178/300 trajectories; 95% CI, 51.0–67.6), followed by GPT-5.5 on Pi at 55.3% (166/300; 47.0–63.6), Claude Opus 4.8 on Claude Code at 54.7% (164/300; 45.9–63.4), and Gemini 3.5 Flash on Pi at 51.3% (154/300; 42.9–59.8). These performances overlap substantially, indicating a broad leading tier rather than a decisive winner. More importantly, even the best-performing configuration remained below 60%, leaving more than 40% of task attempts unsolved.

Across providers, model generations, and agent harnesses, no system approached the reliability expected for preclinical decision-making. The leading tier was broad, and even its best member left a large fraction of local pharmacology decisions unsolved.

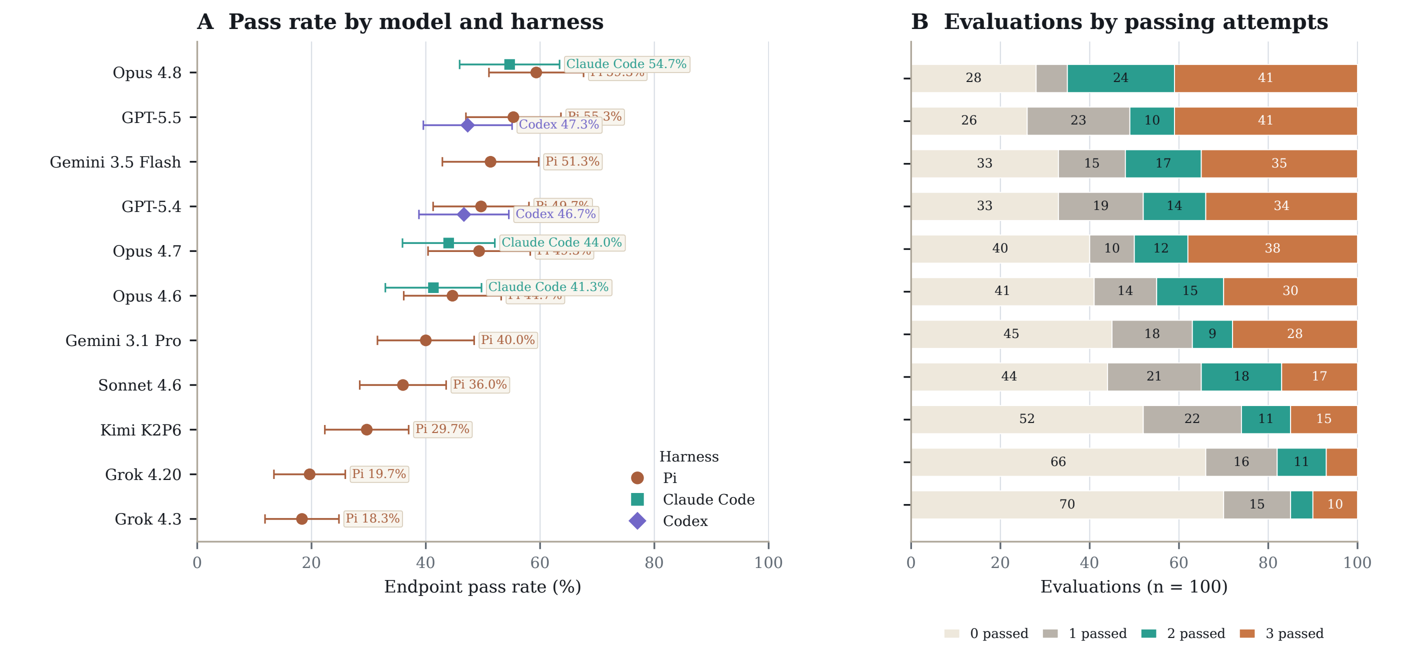


Figure 3: **Pass rates and reliability across model–harness configurations.** (**A**) Endpoint pass rate for each of the 16 model–harness configurations across all 4,800 agent trajectories, with 95% confidence intervals computed over task-level pass rates. Each trajectory was scored pass only when its submitted answer satisfied every task-specific grader. In-panel labels identify the harness for each point; alternate harnesses (Claude Code, Codex) are shown as offset points for the models that ran them. (**B**) Task-level reliability for the Pi harness, partitioning the 100 evaluations for each model by how many of the three independent rounds passed.

Model / harness	Endpoint pass rate	$\geq 1/3$	$\geq 2/3$	3/3
Claude Opus 4.8 / Pi	59.3% (178/300; 51.1–67.6)	72/100	65/100	41/100
GPT-5.5 / Pi	55.3% (166/300; 47.0–63.6)	74/100	51/100	41/100
Claude Opus 4.8 / Claude Code	54.7% (164/300; 45.9–63.4)	65/100	59/100	40/100
Gemini 3.5 Flash / Pi	51.3% (154/300; 42.9–59.8)	67/100	52/100	35/100
GPT-5.4 / Pi	49.7% (149/300; 41.3–58.0)	67/100	48/100	34/100
Claude Opus 4.7 / Pi	49.3% (148/300; 40.4–58.3)	60/100	50/100	38/100
GPT-5.5 / OpenAI Codex	47.3% (142/300; 39.6–55.1)	70/100	46/100	26/100
GPT-5.4 / OpenAI Codex	46.7% (140/300; 38.8–54.5)	68/100	46/100	26/100
Claude Opus 4.6 / Pi	44.7% (134/300; 36.1–53.2)	59/100	45/100	30/100
Claude Opus 4.7 / Claude Code	44.0% (132/300; 35.9–52.1)	62/100	45/100	25/100
Claude Opus 4.6 / Claude Code	41.3% (124/300; 32.9–49.7)	55/100	43/100	26/100
Gemini 3.1 Pro / Pi	40.0% (120/300; 31.5–48.5)	55/100	37/100	28/100
Claude Sonnet 4.6 / Pi	36.0% (108/300; 28.4–43.6)	56/100	35/100	17/100
Kimi K2P6 / Pi	29.7% (89/300; 22.3–37.0)	48/100	26/100	15/100
Grok 4.20 reasoning / Pi	19.7% (59/300; 13.4–25.9)	34/100	18/100	7/100
Grok 4.3 / Pi	18.3% (55/300; 11.9–24.8)	30/100	15/100	10/100

Table 1: **Pass rates by model–harness configuration.** Pass rates are reported as percentages, with the numerator and denominator shown as passing trajectories out of 300 total trajectories per configuration, together with 95% confidence intervals. Confidence intervals were computed across task-level mean pass rates over the 100 preclinical pharmacology tasks. The final three columns report the number of tasks, out of 100, for which at least one, at least two, or all three independent attempts passed.

Most failures map to a compact failure-mode taxonomy

To understand why agents fail, we manually reviewed 1,834 failing Pi-harness trajectories and labeled each against the evaluation's documented scientific intent. We classify each failure along two complementary axes: the *scientific error*, which captures what is wrong with the answer, and the *behavioral cause*, which captures how the model arrived there (Figure 4).

The scientific axis comprises five recurring error families (Figure 4A, restricted to the $n = 1{,}436$ failures that contain a scientific error). The largest, flawed method, resolves into two main subcategories and a small residual category. A *skipped-QC* error (17%) omits a required preprocessing or quality-control step: in one case, a model aggregated clustered regularly interspaced short palindromic repeat (CRISPR) guides to gene-level scores without first dropping guides that map to multiple loci, so paralog-family artifacts flooded the essential-gene list. A *wrong-reference-frame* error (21%) runs the analysis correctly but uses an incorrect reference statistic. The small *wrong-assay-layer* tail (2%) uses the wrong component of the data. On a proteasome-kinetics task asking which compound degrades substrate faster, the answer lives in the ubiquitin-remnant layer of a dual-readout mass-spectrometry run. A model that uses the phosphoproteome data reaches a different conclusion.

Beyond flawed method, four families remain. A *calibration* error (31%) draws a cutoff a domain expert would set differently. For instance, in one competition-binding panel, models carried forward all 33 nonzero kinases when only the 8 strong binders above a clear affinity gap were supported. A *perception* error (10%) misreads a raw image, such as counting apoptotic debris as viable organoids or mis-segmenting two-channel colocalization. *Prior over data* (10%) answers from textbook or famous biology against the staged evidence: one model computed the strongest data-supported biomarker—a vanadium compound correlated with SLC26A2 at $r = -0.48$—then discarded it for the textbook elesclomol-FDX1 cuproptosis pair. *Over-reading* (10%) states a stronger claim than the data supports, such as reporting tumor regression from a sub-baseline median when the per-animal measurements show only growth arrest. Together, method and calibration account for 71% of scientific errors.

The behavioral axis records how each failure happened, across all $n = 1{,}834$ runs (Figure 4B). Two-thirds (67%) are *engaged, analytically wrong*: the model ran a relevant analysis but made an analytical error. The remaining third splits into one evidence-handling failure and three process failures. *Prior-driven override* (6%), the only evidence-handling case, engages the data and often computes the supporting result, then defers to a textbook or famous-biology prior. *Weak commitment* (9%) performs the right analysis but submits a final answer that does not match it, a mismatch that appears even in the strongest models and points to a missing final-answer check. *Abandoned-correct* (12%) has a passing result within reach and then commits to a worse, more canonical one such as a published figure or standard global method. Like prior-driven override, it occurs most often in the longest runs. *Under-investment* (6%) reaches an answer with little tool use, submitting a plausible decision before running the per-dose computation or QC the task requires.

The two axes distribute differently across models (Figure 4C,D). The behavioral mix varies by model. Under-investment is concentrated in the two weakest models, at 32% of Grok 4.3's failures and 17% of Grok 4.20 reasoning's against 0–1% for the others. Abandoned-correct is spread across stronger models at 7–17%, highest in GPT-5.5 and Gemini 3.1 Pro (17% each). The scientific-error mix is nearly constant, with method and calibration accounting for 58–77% of every model's scientific errors regardless of pass rate.

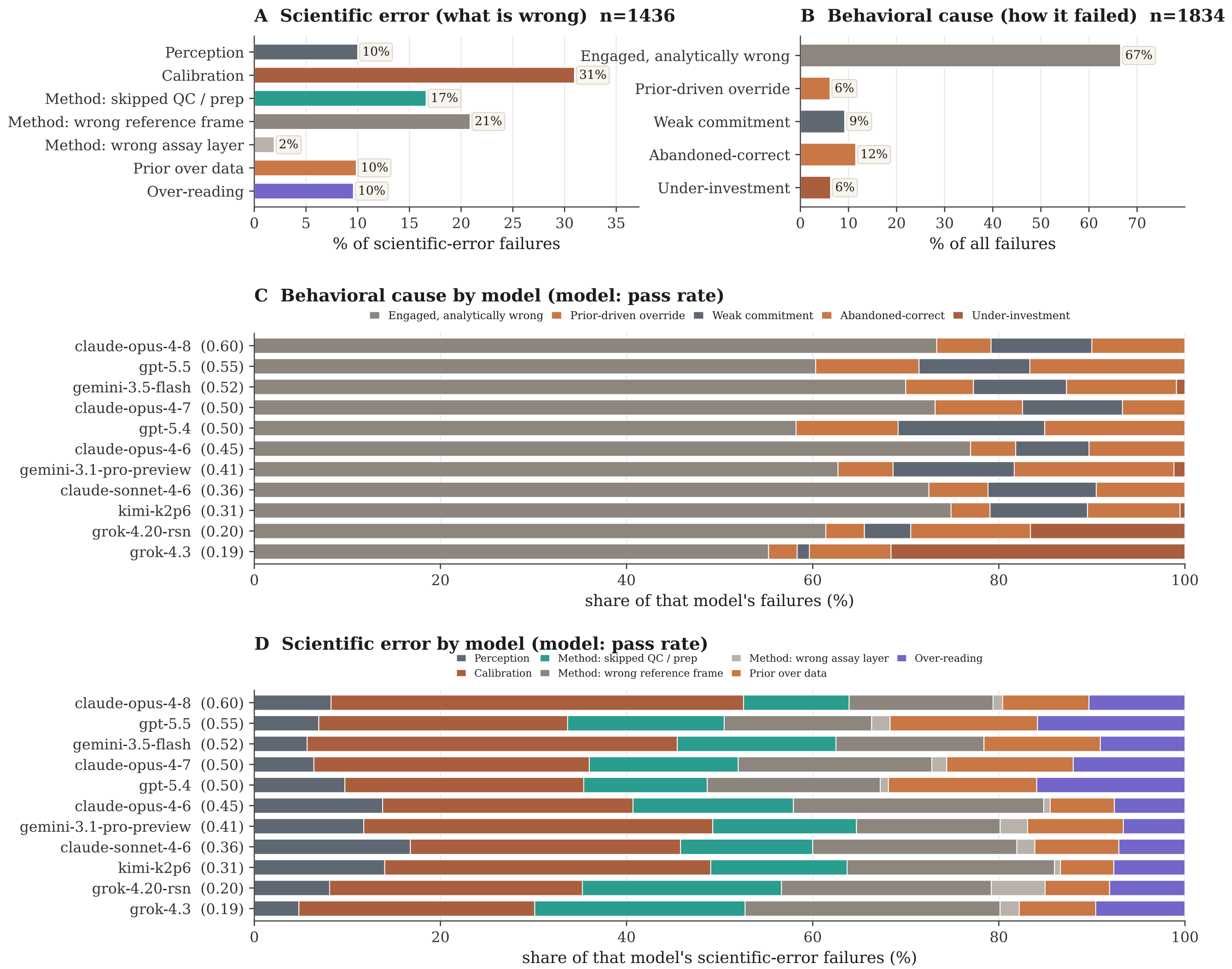


Figure 4: A two-axis failure taxonomy. The 1,834 reviewed failing Pi-harness trajectories, labeled against each evaluation's scientific intent and classified along two complementary axes. (**A**) The scientific-error axis (what is wrong with the answer) is dominated by method and calibration errors (71% combined), reflecting analysis and calibration choices rather than recall of known drug mechanisms; the method family resolves into a skipped-QC/preprocessing sub-mode and a larger wrong-reference-frame sub-mode. (**B**) The behavioral-cause axis (how the model failed) shows that a third of failures are non-analytical: a prior-driven override of the model's own evidence, plus three process breakdowns (weak commitment, abandoned-correct, under-investment). (**C**) The behavioral mix differs by model: under-investment dominates only the weakest model, while the strongest models fail almost entirely through engaged, analytically wrong reasoning. (**D**) The scientific-error mix, by contrast, is largely model-invariant: method and calibration errors dominate every model regardless of overall accuracy, so the *kinds* of scientific mistakes are shared across stronger systems even as their rate falls (each bar normalized within that model's scientific-error failures). A scientific-error × behavioral-cause cross-tabulation is provided in the supplement (`fig_fm_crosstab`).

The dominant failure mode shifts across program stages

Pass rates range from 27% at the hardest stage to 55% at the easiest stage (Figure 5). Stage-level examples show different failure patterns across the program-stage axis. Below, parenthetical percentages give the fraction of passing runs for representative tasks.

Overall, the ordering of models at each stage closely mirrors their overall performance (Figure 5C; mean Spearman $\rho = 0.80$ across stages, ranging from 0.56 at S7 to 0.92 at S8), with the two Grok models ranking lowest at every stage. Three patterns stand out. The two highest cells in the matrix are Gemini 3.5 Flash and GPT-5.5 at drug response, at 87% (95% CI 70–95, $n = 30$) and 86% (69–95, $n = 29$), well above their own overall rates of 52% and 55%. The clearest stage effect is within a single model. Gemini 3.5 Flash falls from 87% at S3 to 28% (16–44, $n = 36$) at causal target validation. The widest within-family spread is at hit screening, where Claude Opus 4.8 scores 50% (31–69, $n = 24$) and the older Claude Opus 4.6 scores 12% (4–31, $n = 24$).

Primary screening and hit prioritization (27%) is the hardest stage. Models are competent at the mechanical steps, such as robustly collapsing noisy replicates (`PRISM05`, 58%). They fail by choosing the wrong reference frame for QC. On outlier-pool QC (`PRISM01`, 18%) they flag whole well-by-pool groups by an aggregate median residual and exclude most of the cytotoxic positive-control wells, discarding real biology as a technical artifact. On an oncology drugs repurposing prioritization task (`PRISM_S2_11`, 0%) they rank by breadth of killing and nominate a pan-cytotoxic compound rather than a selective hit.

Drug response (55%) is one of the easier stages but exposes a consistent weakness in unsupervised mechanism recovery. Models do well on direct evidence checks, such as separating a causal mediator from a correlated biomarker (`PRISM_S4_06`, 76%). They fail when asked to cluster compounds by mechanism from their response profiles. Across the embedding tasks they use a potency-dominated representation, linear principal component analysis (PCA), or a Euclidean metric on raw profiles, so the leading component encodes potency and the intended drug class never separates from the rest. The same error recurs across statin collapse (`PRISM_UMAP_01`, 12%), oncology MoA recovery (`PRISM_UMAP_04`, 48%), and glucocorticoid substructure (`PRISM_UMAP_03`, 69%), so this is a repeated stage-level pattern rather than a single-task failure.

Causal target validation (32%) is the second-hardest stage and the one that most rewards data hygiene and statistical rigor. Models recover genuine dependencies from a clean screen (`HIT_05` replicate discordance, 69%). They routinely skip the library hygiene that CRISPR screens require. On multi-target guide contamination (`GG_01`, 16%) they aggregate guides to gene level without dropping multi-locus single-guide RNAs (sgRNAs), so paralog-family false positives flood the dependency list. On the selective-set call (`SEL_01`, 6%) they ship thousands of genes behind a count-only filter with no effect-size floor.

Mechanism of action and pharmacodynamics (*MoA/PD; 37%*) is the largest stage, with 21 evaluations. Models reliably recover known program structure, such as assigning cells to the correct cell-cycle phase (`clust_04`, 84%). Their recurring failure is over-reaching on the mechanism, treating marginal or partial evidence as a strong, broad effect. On the antiinfective task (`ANTIINF_01`, 0%) they read near-noise pathway scores ($p \approx 0.05$) as genuinely engaged mechanisms. On the mechanistic target of rapamycin (mTOR) fingerprint (`TXBX_DM_v15E`, 6%) they correctly call mTOR the primary target but then count phosphosites from too broad a substrate panel, sweeping in adjacent AKT-axis sites instead of restricting to the canonical mTOR set, which inflates the affected-site count to almost twice what the data supports (59 versus 31).

Compound-target engagement (43%) asks whether a compound truly hits its target and how selectively, and the bottleneck is calibration, not detection (67% of the stage's scientific errors). Models detect the binding signal reliably, for example confirming a clean direct target by thermal shift (`thermal_shift_shared_direct_target`, 91%). What they lack is a principled cutoff for how much engagement counts, so they err in both directions. On a kinase-binding panel (`TXBX_KB_v11_DA1`, 0%) they accept every weak binder as a real hit, and on a control task (`CTRL01`, 21%) they invent a stringency cutoff the data never justified and discard five genuinely engaged targets.

Developability and safety (43%) depends on judging a signal against a domain-specific reference. Models often make a confident call when the standard is well established in the literature, such as a hepatotoxicity task (`TXBX_TG_v11_A`, 91%). On the human ether-a-go-go-related gene (hERG) safety-margin task (`de_14`) they size the margin against total drug concentration rather than the free, unbound fraction that actually reaches the target, overstating the safety window. The same behavior appears on a cardiotoxicity task (`de_12`, 0%). The pathways most associated with a three-level toxicity-concern score must be found with a rank test across all three levels, but models collapse the score to a yes/no label and rank pathways by a binary test, so their top pathways diverge from the correct set and they miss the potassium-channel mechanism the data supports.

Translational efficacy (49%) reveals failures reading microscopy images (image perception is 62% of its scientific errors). In contrast, models do well when the evidence is a table, such as calling a genotype-by-treatment rescue (`HAI2027_MM_FIG6C`, 89%). On a two-marker immunostain (`HAI2027_MM_FIG2E`, 42%) they read faint, near-vehicle single-agent staining as a real signal and credit each monotherapy with activity, endorsing a single-agent program when only the drug combination is active. The organoid viability readouts are the hardest tasks in the benchmark (as low as 6%). When apoptotic debris is counted as living organoids (`organoid_moa_cytotoxic`), the mechanism call inverts, and a compound that is in fact killing the tumor cells is reported as inert or growth-promoting and advanced on that false basis.

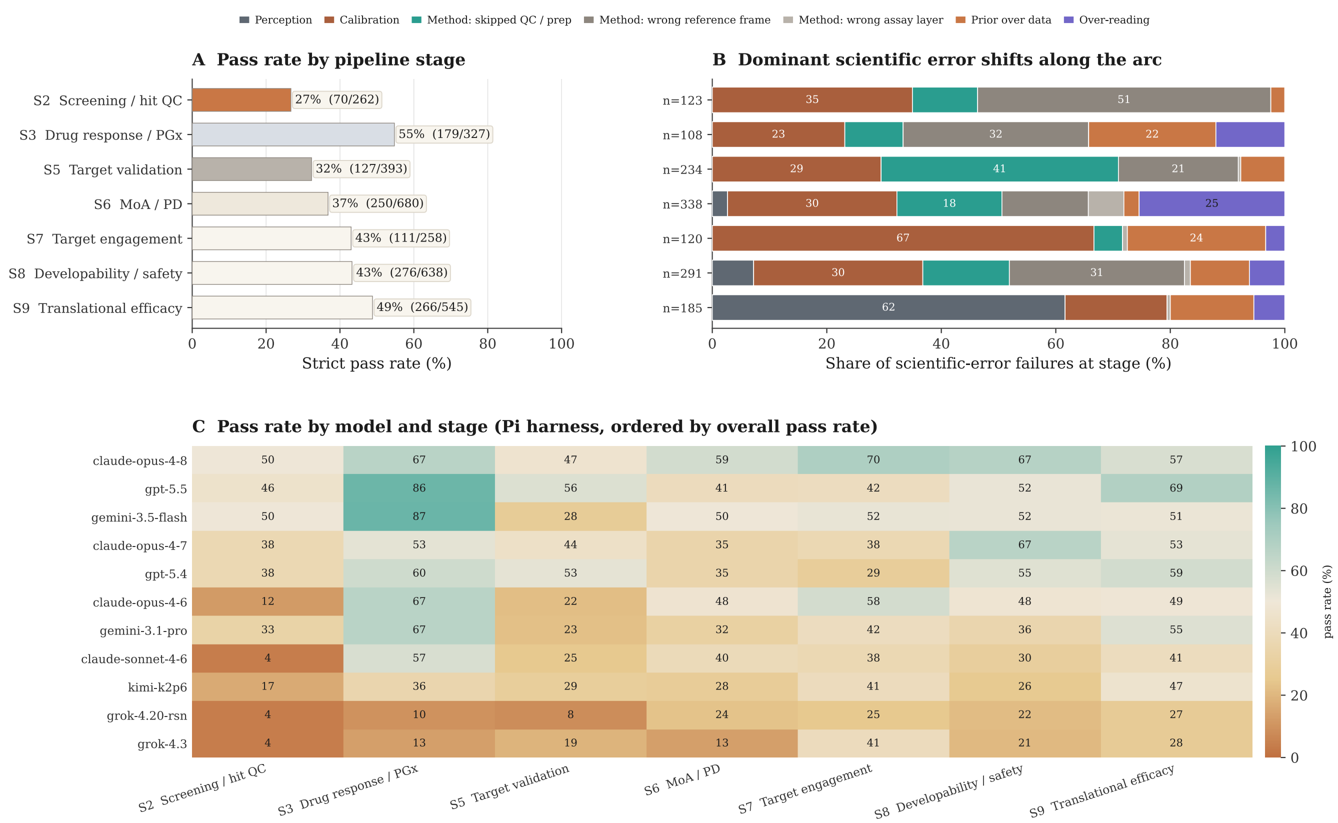


Figure 5: The dominant failure mode shifts across program stages. Failing Pi-harness runs placed along the program-stage axis. S1 disease/model characterization and S4 human genetic target support are omitted, since S1 is backed by too few evaluations and S4 is reserved for future benchmarking work. (**A**) Endpoint pass rate by stage. The two lowest are the early data-handling stages S2 (screening/QC) and S5 (causal target validation). (**B**) Scientific-error composition by stage, restricted to failures carrying a scientific error and using the same seven categories as Figure 4A. The dominant error changes by stage, from wrong reference frame at screening (S2) to skipped QC at target validation (S5), calibration at target engagement (S7), and perception at the image-heavy translational stage (S9), while calibration persists as a thread at every stage. (**C**) Endpoint pass rate for every model at each stage, ordered by overall pass rate. The overall ordering is largely preserved across stages, with two clear rearrangements: Gemini 3.5 Flash and GPT-5.5 lead at drug response (S3), and Gemini 3.5 Flash drops sharply at causal target validation (S5).

Some task categories remain far harder than others

Tasks become more difficult when they require nuanced biological reasoning and judgment. Grouping the 100 evaluations by task type and restricting to the 14 categories with at least four evaluations (86 of the 100 evaluations), the mean Pi-harness pass rate across the 11 models spans a 4.1-fold range, from 70% on exposure and PK tasks to 19% on hit prioritization and 17% on cheminformatics (Figure 6). Performance separates by the structure of the required answer rather than by subject matter. The four highest categories, exposure-PK (70%), biomarker discovery (56%), safety assessment (52%), and target engagement (51%), each resolve to a determinate target and average 57%. In each, the answer is either a number computed and compared against a threshold or a single entity identified by the evidence (a biomarker, or a binary safety or engagement call). The four lowest categories, target dependency (30%), differential response (28%), hit prioritization (19%), and cheminformatics (17%), instead require selecting a calibrated subset from many candidates with no unique answer, and average 24%. These tasks ask which genes are selective dependencies, whether a response is shared or lineage-selective, which screen hits to advance, and which substructures are genuine safety liabilities. The two groups differ by 34 percentage points (2.5-fold), and the ordering matches the failure taxonomy, in which tasks with a definite target are tractable for current systems while those requiring a calibrated cut under uncertainty are not.

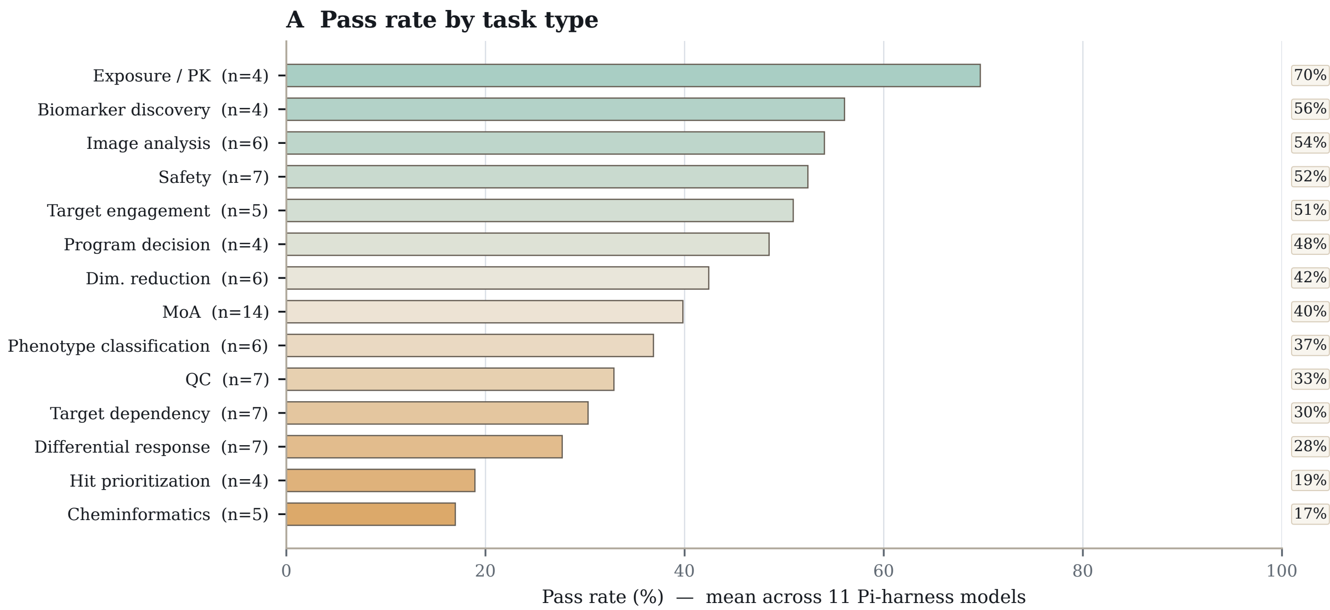


Figure 6: Some task categories remain far harder than others. Pi-harness pass rate by task type (categories with at least four evaluations), averaged across the 11 models. Difficulty tracks the kind of judgment required, from determinate quantitative tasks (exposure-PK) down to open-ended selection tasks (hit prioritization, cheminformatics).

Splitting performance by model and task type (Figure 7) shows that the two effects are largely separable. Within a model (a row of the matrix), overall model strength dominates, with the strongest models (Claude Opus 4.8, GPT-5.5) scoring high across most categories and the weakest (Grok 4.3, Grok 4.20 reasoning) scoring low across all of them. Within a task type (a column), difficulty imposes a ceiling that a stronger model does not overcome. On the two hardest categories, hit prioritization and cheminformatics, no model exceeds 45%, indicating a limit shared across current models.

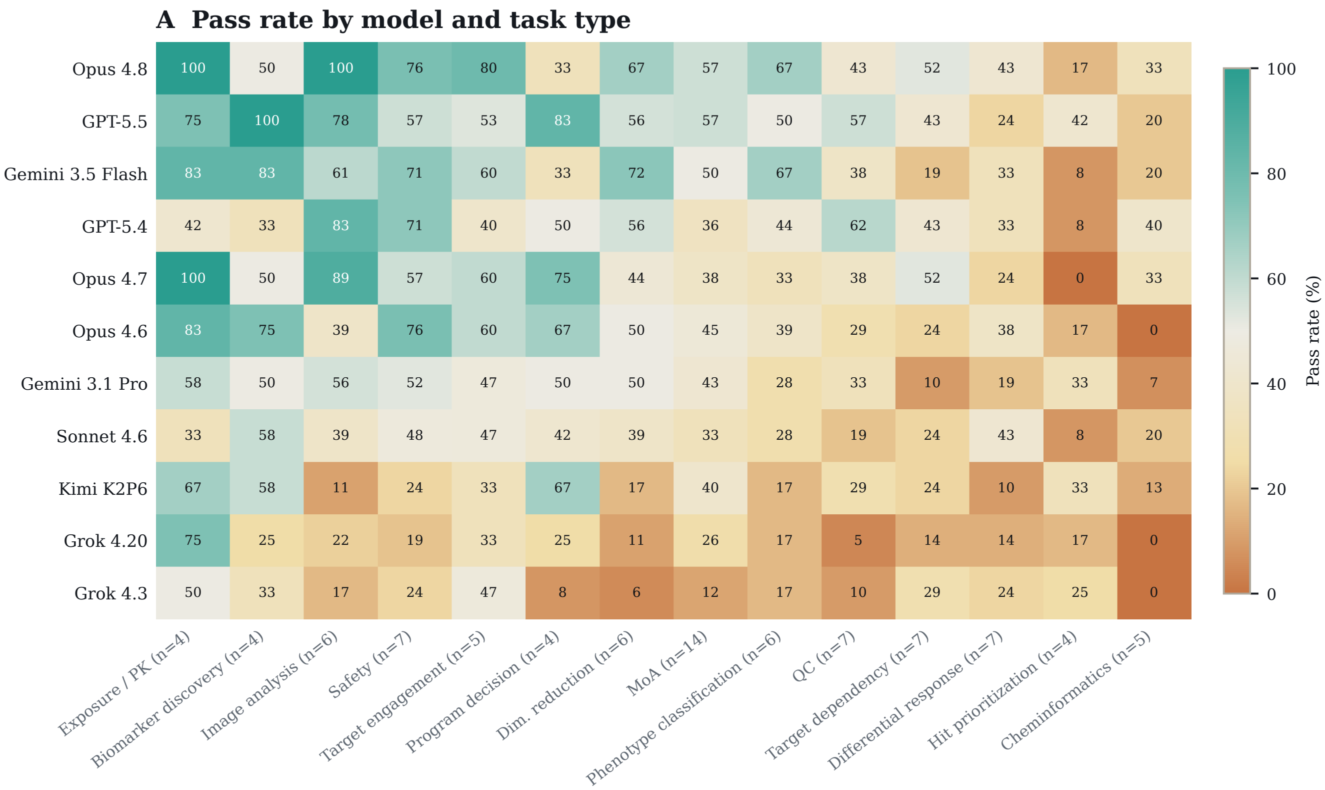


Figure 7: Pass rate by model and task type. Strict Pi-harness pass rate for each of the 11 models (rows, ordered by overall pass rate) across the task categories with at least four evaluations (columns, ordered easiest to hardest). Model quality sets the level across a row, but the hardest categories on the right stay low for every model.

Among strong models, accuracy tracks process discipline rather than effort

Pass rate is only weakly and non-monotonically related to agent cost or steps (Figure 8). Across the 11 models, mean agent steps and endpoint pass rate correlate at Spearman $\rho = 0.54$, dropping to $\rho = 0.32$ when the two low-engagement Grok models are excluded. Grok 4.3 (mean \$0.02, median 6 steps) and Grok 4.20 reasoning (\$0.11, median 11 steps) score lowest, at 18.3% and 19.7%, and under-investment drives 32% and 17% of their failures, against 0–1% for every other model (Figure 4C). Above the floor, effort and accuracy decouple. Gemini 3.5 Flash takes the most steps (median 43, $\approx 2.5\times$ Claude Opus 4.8's 17) but reaches only 51.3%, below Claude Opus 4.8's 59.3%. GPT-5.5 beats GPT-5.4 (55.3% vs. 49.7%) with fewer steps (median 15 vs. 24). Claude Opus 4.6 reaches 44.7% at a median of 10 steps, above Claude Sonnet 4.6 (36.0%) and Kimi K2P6 (29.7%) despite their higher step counts.

Strong models differ mainly in the composition of their failures (Figure 4C). The top Claude models fail mostly through engaged-but-incorrect analysis. Analytical errors account for 73–77% of their failures, leaving 23–27% non-analytical (Claude Opus 4.6 lowest at 23.0% of $n = 165$ failures, Claude Opus 4.7 at 26.8%, Claude Opus 4.8 at 26.7%). The GPT models reach comparable accuracy with more finalization-related errors. Weak commitment (GPT-5.5 12%, GPT-5.4 16%) and abandoned-correct answers (17% and 15%) raise their non-analytical share to 39.7% (GPT-5.5, $n = 126$) and 41.8% (GPT-5.4, $n = 146$). Their remaining errors therefore often occur after useful analysis: the model fails to verify the read or submit the answer its analysis already supports. Under-investment is effectively absent from every strong model ($\leq$1% of failures) and appears only in the two cheapest.

These quantitative patterns correspond to identifiable behaviors on individual tasks. Successful runs identify the scientific intent, draw a calibrated cut, verify against the data, and commit to one answer. The pattern is clearest on hard tasks, where most runs fail but the strongest still succeed, and each hinges on a concrete drug-program decision. For instance, Bosutinib is an approved BCR-ABL/SRC inhibitor, but a program must credit its activity to the targets it actually engages in the chemoproteomics panel. On this task (32% of runs passing), Claude Opus 4.8 overrode the clinical prior and committed to the non-canonical profile the panel supports, while other models anchored to the label returned the canonical ABL/SRC answer. The crizotinib control (21%) asks whether a separate broad-spectrum control compound justifies dropping genuinely engaged targets in a live-cell NanoBRET target-engagement screen. It does not, and Claude Opus 4.8 kept five engaged targets that other models discarded behind an invented quality-control gate, an error that would have removed real hits from medicinal-chemistry follow-up. The epidermal growth factor receptor (EGFR)/phospho-mitogen-activated protein kinase kinase (MEK) task (44%) asks whether EGFR productively drives downstream MEK signaling in a BRAF-mutant colorectal-cancer organoid, the call that decides whether the compound advances as on-pathway. GPT-5.5 quantified the co-positive cell share in code rather than calling it from total marker abundance, reaching the supported conclusion.

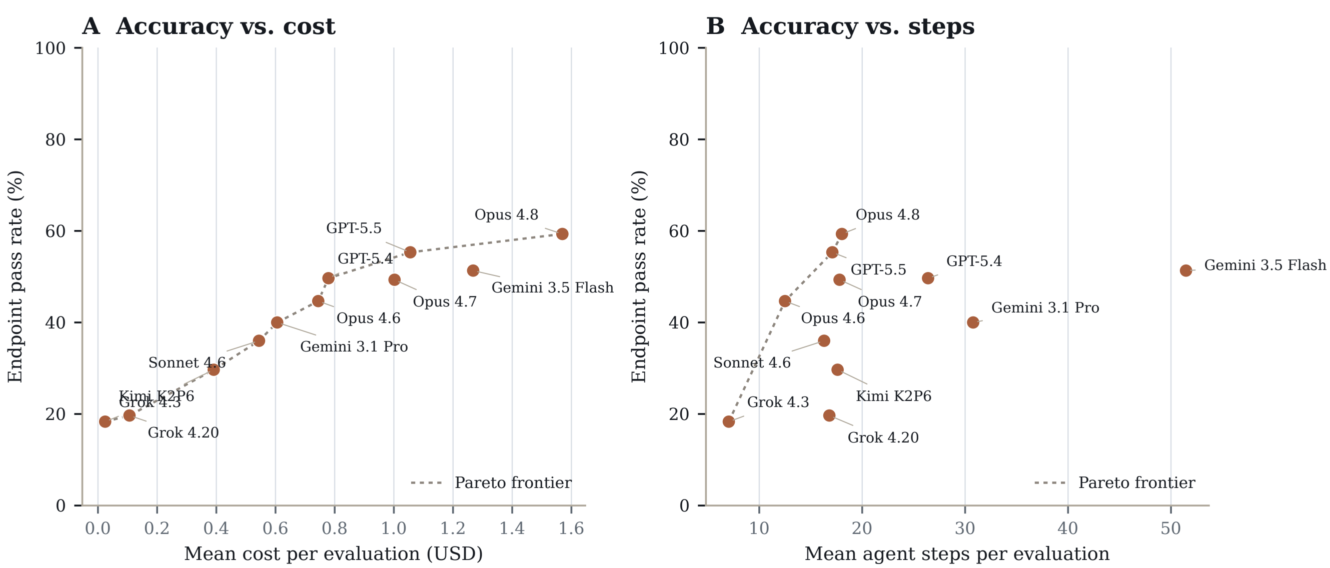


Figure 8: Spending more does not buy accuracy. Strict Pi-harness pass rate against mean cost (A) and mean agent steps (B) per evaluation. The dotted line is the Pareto frontier. The cheapest, fewest-step models score lowest, but above a modest budget effort and accuracy decouple. The highest-step model (Gemini 3.5 Flash) is not the most accurate, and Claude Opus 4.8 leads at moderate effort.

Failures translate directly into wrong go/no-go decisions

Some evaluations closely resemble real program decisions, giving the agent several assay readouts at once and asking for a single advance, hold, or prioritize call. The current suite includes seven such advancement decisions; across 230 runs, models pass only 35% of them (Figure 9). Models perform well when the data are clean and consistent, but struggle on more challenging cases, either advancing weak candidates or discarding strong ones.

Clean correct. The organoid program go/no-go is the clean case (Figure 9). Given a primary screen, a mechanism image, a toxicology counterscreen, and DMPK exposure for eight blinded compounds, the supported action is to advance the one compound that shows the desired cytostatic mechanism together with an adequate safety margin and exposure coverage, and most models do (82%).

Wrong synthesis. On some decision tasks, the agent must assemble a set of models with convergent support across several endpoints, but models rank on a single metric (27%), so the program's next experiments are built on wrong assumptions.

False-go. On the safety advancement task the supported call is that no candidate moves forward, because none pairs activity with healthy-tissue sparing, yet models advance one on activity alone or on a relative tumor-versus-healthy index (36%), putting an unsafe compound into the pipeline.

False no-go. The reverse error also appears. On the long-term combination task, models read a strong colony-stain reduction as a complete kill and drop a model the combination still supports (`HAI2027_FIG1_DUR01`, 48%).

We found strong overall benchmark scores do not predict making the better program decisions. The six multi-evidence advancement decisions with complete Pi-harness model comparisons yield an order almost unrelated to overall accuracy (Spearman $\rho = 0.08$; Figure 10A). Claude Opus 4.8, first overall at 60%, ranks last on these decisions at 22% (4/18). It clears both single-compound calls, advancing the one qualifying organoid compound (2/3) and holding when no candidate is safe (2/3), but it fails all four decisions that instead ask which preclinical models (cell lines and patient-derived xenograft (PDX) models) should anchor the next program stage (the single-agent, priority, long-term, and decision-gate sets in Figure 10B), passing none of them (0/3 each) because it over-ranks on a single metric or carries too many candidates forward. GPT-5.5 is the only model strong on both kinds of task (15/18, 83%), committing to a defensible prioritized set rather than a full ranking. Kimi K2P6, ninth overall at 31%, rises to second here (12/17, 71%): its careful per-model computation pays off when the task rewards naming the supported models, though it still fails the restraint call by advancing a candidate on the no-go safety task (0/3). The two Grok models stay at the bottom (5/18 each) by answering with little of the multi-evidence work the decision requires.

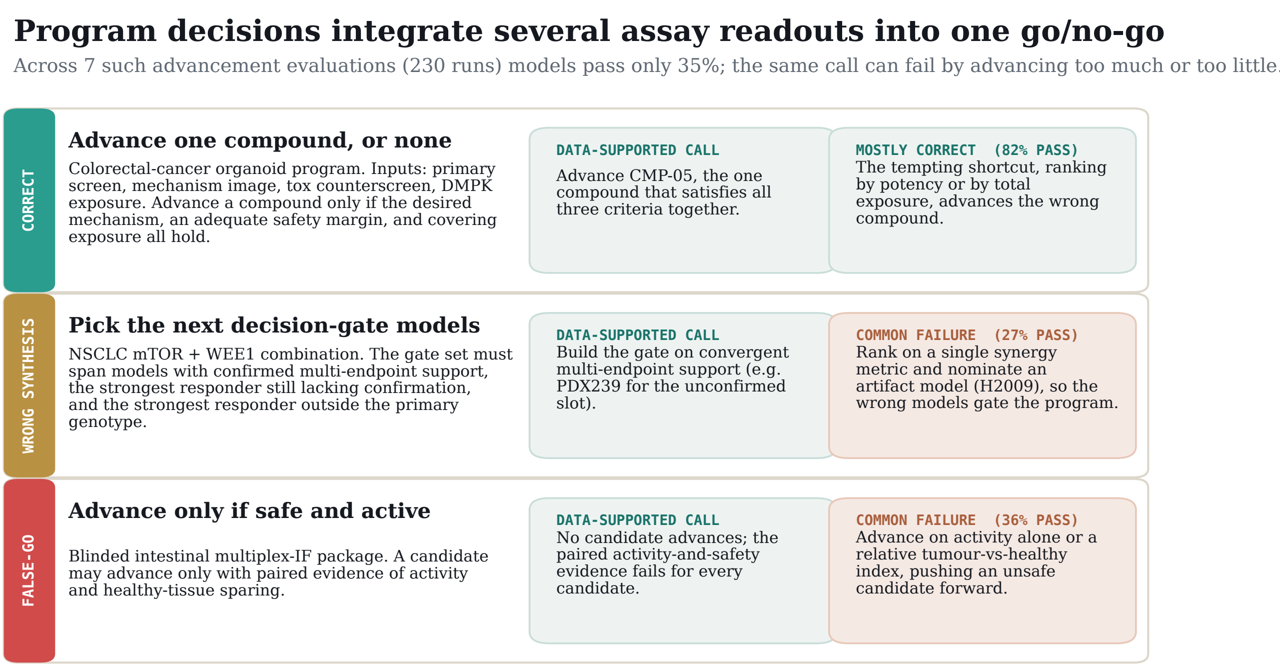
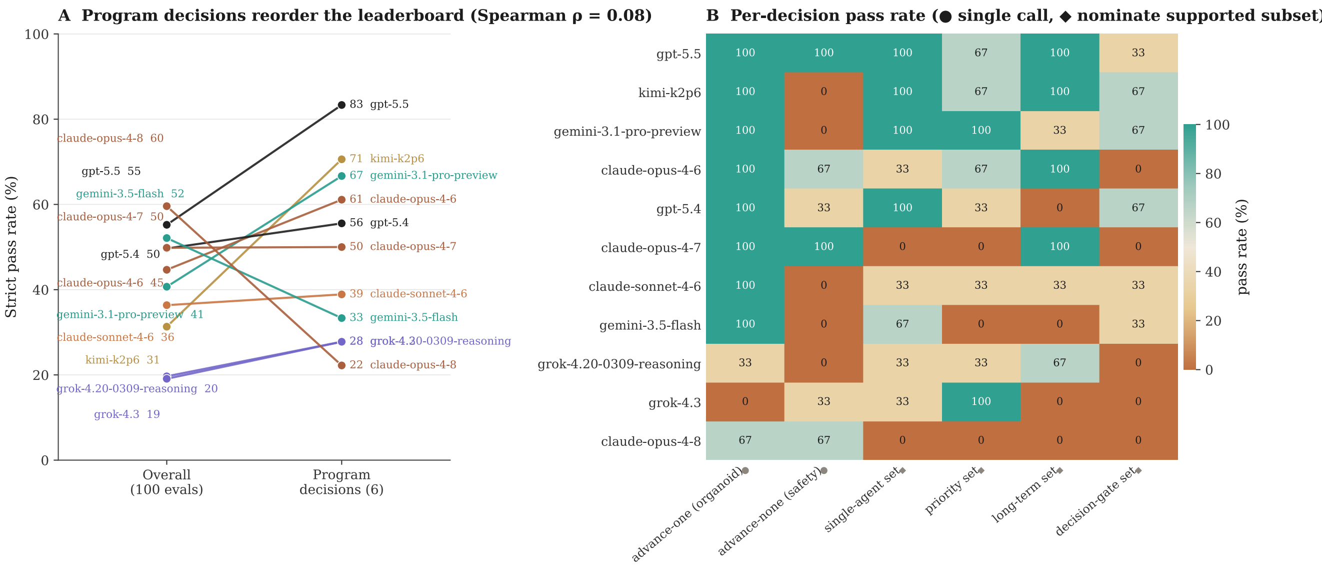


Figure 9: **Program decisions turn analysis errors into wrong go/no-go calls.** Three multi-evidence advancement evaluations, each giving the agent several assay readouts and asking for one advance, hold, or prioritize call. For each we show the evidence and the question, the data-supported call, and the common model error with its program consequence. The organoid go/no-go is passed by most models (82%); the decision-gate and safety-advancement tasks fail in opposite directions, advancing the wrong models (27%) or an unsafe candidate (36%). NSCLC denotes non-small-cell lung cancer.



	advance-one (organoid)	advance-none (safety)	single-agent set	priority set	long-term set	decision-gate set
gpt-5.5	100	100	100	67	100	33
kimi-k2p6	100	0	100	67	100	67
gemini-3.1-pro-preview	100	0	100	100	33	67
claude-opus-4-6	100	67	33	67	100	0
gpt-5.4	100	33	100	33	0	67
claude-opus-4-7	100	100	0	0	100	0
claude-sonnet-4-6	100	0	33	33	33	33
gemini-3.5-flash	100	0	67	0	0	33
grok-4.20-0309-reasoning	33	0	33	33	67	0
grok-4.3	0	33	33	100	0	0
claude-opus-4-8	67	67	0	0	0	0

Figure 10: **Aggregate accuracy does not predict program-decision competence.** Pass rates on the six multi-evidence advancement decisions with complete Pi-harness model comparisons (up to 18 runs per model). **(A)** Endpoint pass rate on all 100 evaluations versus on the six program decisions; the integrative go/no-go calls reorder the leaderboard almost completely (Spearman $\rho = 0.08$), with the overall-best model, Claude Opus 4.8, falling to last and Kimi K2P6 rising from ninth to second. **(B)** Per-decision pass rate by model. The two single-compound calls (advance one, or hold; marked •) are passed broadly, but the four tasks that require nominating the supported subset of models (marked with a diamond) separate the field; Claude Opus 4.8 fails all four.

The agent harness has a surprising impact on model performance

Each harness used the same model checkpoint, task text, and answer-submission protocol, but pass rate still varied by harness. In matched comparisons within the same model family, Pi outperformed Claude Code by 4.4 percentage points (95% CI: 1.2–7.8) on the shared Claude Opus models and outperformed OpenAI Codex by 5.5 percentage points (95% CI: 1.2–9.8) on the shared GPT models. Pi scored higher on every shared model, not only on average.

The size of this difference is comparable to a model-generation upgrade. Moving from Claude Opus 4.6 to Claude Opus 4.7 improved Pi performance by 4.6 points, similar to the harness effect. The Pi advantage was also similar on image-based and non-image tasks, so it is not explained only by visual access. Claude Code rendered figures to the model in 100% of image-bearing tasks, compared with 84% for Pi; OpenAI Codex, which lacks vision capabilities, never rendered figures.

The remaining differences point to implementation details. Pi's lean, three-tool loop failed on 0.6% of runs, compared with 1.8% for Claude Code, and reached answers in fewer steps. OpenAI Codex was also penalized on figure-dependent tasks because it did not render figures. These effects are small in absolute terms and have wide confidence intervals, but they show that reported scores are properties of the model–harness configuration rather than the model alone.

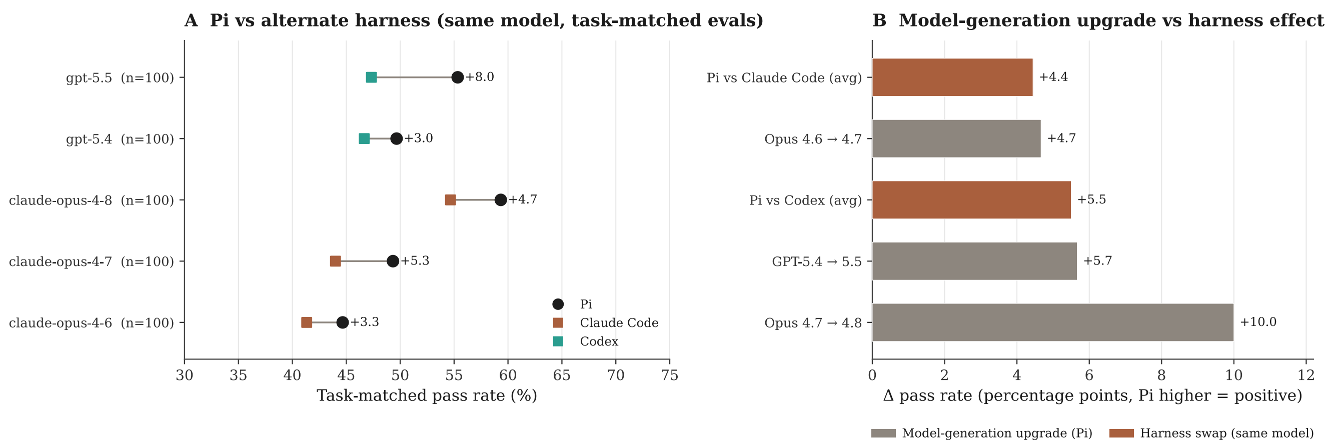


Figure 11: Harness choice is as consequential as a model-generation upgrade. (**A**) Task-matched pass rate on Pi versus the alternate harness for each model pair. Points are matched on the shared evaluation set; the labeled gap is the within-model Pi advantage in percentage points. (**B**) Head-to-head comparison of the Pi advantage over an alternate harness (terracotta) against the gain from a model-generation upgrade on Pi (gray). A harness substitution shifts pass rate by a margin comparable to upgrading one model generation, so reported scores should be interpreted as a property of the model–harness configuration, not the model alone.

Discussion

TxBench-PP measures a narrow but important part of drug discovery: whether an agent can turn preclinical pharmacology evidence into a defensible local decision. This scope is intentionally smaller than end-to-end therapeutic discovery. It is a small-molecule task-group benchmark within the broader TherapeuticsBench roadmap, not a claim to cover every stage or modality. It does not yet test human genetic target support, full disease-model selection, broad clinical translation, or non-small-molecule modalities. The narrower scope makes the benchmark verifiable while still covering decisions that directly affect compound advancement.

The results support this distinction. Claude Opus 4.8 / Pi was the leading configuration, passing 59.3% of endpoint attempts (178/300; 95% CI, 51.1–67.6), but still failed more than two of every five attempts. Reviewed failures were not mainly blank answers or missing execution. Most were data-engaged analyses that made the wrong method, calibration, perception, or prior-versus-data choice. The program-decision tasks show the practical consequence: local analysis errors can advance weak or unsafe candidates and discard supported ones.

Future versions can extend the benchmark upstream to target rationale and disease-model characterization, sideways to broader pharmacogenomics, and across modalities to antibodies, antibody-drug conjugates (ADCs), degraders, oligonucleotides, cell therapies, and gene therapies. A complementary long-horizon TherapeuticsBench track can follow program stories across discovery, development, and translation. For the current paper, the clean claim should remain: TxBench-PP is a verifiable task-group benchmark for small-molecule preclinical pharmacology decisions.

Methods

Benchmark assembly

Benchmark construction is described in the Benchmark Construction and Benchmark Anatomy sections. Briefly, we assembled 100 evaluations from practical small-molecule preclinical pharmacology workflow states. Candidate evaluations were retained when the target decision could be recovered from the supplied data artifacts, expressed through a constrained final answer, and graded deterministically. Tasks were revised or removed when the prompt over-specified the method, when the answer could be recovered by an obvious shortcut, when plausible analysis choices produced incompatible decisions, or when the grader could not distinguish the supported pharmacology conclusion from a plausible but unsupported answer.

Each retained evaluation includes a task prompt, workflow data artifacts, metadata needed to interpret the files, a structured answer schema, a hidden deterministic grader, and tags for program stage, assay type, and task structure. Evaluation notes record the intended empirical decision, known analysis traps, and decision-blocking gaps used for later trajectory interpretation. These notes and failure annotations are not shown to agents.

Agent runs

We evaluated 16 model–harness configurations across Pi, Claude Code, and OpenAI Codex. Pi was run with Claude Opus 4.6, Claude Opus 4.7, Claude Opus 4.8, Claude Sonnet 4.6, GPT-5.4, GPT-5.5, Gemini 3.1 Pro, Gemini 3.5 Flash, Kimi K2P6, Grok 4.20 reasoning, and Grok 4.3. Claude Code was run with Claude Opus 4.6, Claude Opus 4.7, and Claude Opus 4.8. OpenAI Codex was run with GPT-5.4 and GPT-5.5.

Each configuration was run on the same 100 evaluations with three independent attempts per evaluation, yielding 300 trajectories per configuration and 4,800 trajectories overall. Runs used the packaged task context, the supplied data artifacts, a coding environment for file inspection and analysis, and the same structured answer-submission protocol. For each run we recorded the final answer, grader output, trajectory, result log, duration, step count, and available cost metadata.

Failed, timed-out, incomplete, malformed, or unparsable runs were retained in the denominator and counted as nonpassing unless the hidden grader emitted a literal boolean passing result. The result parser treated missing pass fields or non-boolean pass fields as failures.

Endpoint grading

The primary benchmark score is deterministic endpoint grading of the final structured answer. Graders parse the submitted answer and compare the requested fields against task-specific criteria. Depending on the evaluation, these criteria include exact or normalized label matches, numerical tolerances, rank-order checks, categorical choices, and all-of field comparisons. A run passes only when all required task-specific checks pass. Missing fields, invalid JSON, answers outside the expected schema, and answers that cannot be parsed by the grader are counted as failures.

We also report task-level robustness counts for each model–harness configuration: the number of evaluations for which at least one, at least two, or all three replicate attempts passed. These counts are diagnostic and do not replace the endpoint pass rate.

Result aggregation and statistical analysis

For each model–harness configuration and evaluation, the three replicate passes were averaged to produce an evaluation-level pass score in $\{0, 1/3, 2/3, 1\}$. The primary endpoint pass rate is the mean of these 100 evaluation-level scores. We report raw pass counts for interpretability, but compute confidence intervals over evaluations rather than over individual trajectories, because replicate attempts on the same evaluation are not independent samples of the benchmark. Unless otherwise stated, 95% confidence intervals are Student t intervals over evaluation-level mean pass scores.

Stage, assay, and task-type analyses use the same evaluation-level aggregation within each metadata-defined subset. Task-type plots are restricted to categories with at least four evaluations. Cost, duration, and step summaries are averaged first across the three replicate attempts for each evaluation and then across evaluations. Cost analyses use the recorded total-cost metadata when available; for Anthropic model runs without a total-cost field, cost was computed from recorded token usage and the same model-specific price table used for aggregation.

Harness comparisons are matched by model and evaluation. The Claude-family comparison uses Claude Opus 4.6, Claude Opus 4.7, and Claude Opus 4.8 on Pi and Claude Code. The GPT-family comparison uses GPT-5.4 and GPT-5.5 on Pi and OpenAI Codex. Program-decision analyses use the curated subset of evaluations that ask for an advance, hold, prioritize, or related preclinical program decision from multiple evidence streams; the model-comparison panel uses the six such decisions with complete Pi-harness comparisons across the 11 Pi models.

Manual trajectory review

Manual trajectory inspection was used during benchmark construction and for post hoc failure analysis. During construction, trajectories were inspected to identify prompts that over-specified the method, shortcut-solvable tasks, unstable answer surfaces, and graders that failed to separate supported answers from plausible but unsupported ones.

For the failure taxonomy, reviewers labeled 1,834 failing Pi-harness trajectories against the evaluation's documented scientific intent. Each reviewed failure received a behavioral-cause label describing how the model failed. Failures with an assignable scientific issue also received a scientific-error label describing what was wrong with the submitted answer. These labels were used only for diagnostic analysis after endpoint grading.

Data availability

The manuscript source, figure-generation scripts, and derived result summaries are maintained in the TxBench repository. Benchmark task definitions, graders, available trajectories, and redistributable data artifacts will be released with the public TxBench-PP benchmark package. Source assay data are distributed or linked according to the original data licenses and access terms.

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